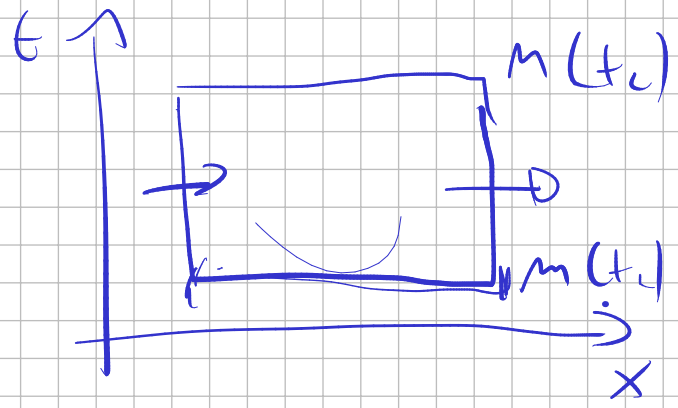
- EQUAZIONI ALLE DERIVATE PARZIALI



$$\partial_t \rho + \partial_x (f(\rho)) = 0$$

- TRASPORTO CON $v = c \in \mathbb{R}$ COSTANTE

$$\partial_t \rho + c \cdot \partial_x \rho = 0$$

- PDE STAZIONARIE $\partial_t u = 0$
- PDE NON-STAZIONARIE $\partial_t u \neq 0$

ORDINE $\underbrace{\partial_{xy} u}_{2^\circ \text{ ORDINE}}$ $\underbrace{\partial_x u}_{1^\circ \text{ ORDINE}} \cdot \underbrace{\partial_y u}_{1^\circ \text{ ORDINE}}$

$$P(u, g) = a(x,t)\, u + b_0(x,t)\, u_t + b_1(x,t)\, u_{x_1} + \dots$$

$$\dots b_d\, u_{x_d} + c_{\alpha(2,0,\dots,0)}\, u_{tt}$$

$$+ c_{\alpha(1,1,0\dots0)}(x,t)\, u_{tx_1} + \dots$$

$$\gamma_{\alpha(p_0,\dots p_d)}(x,t)\,\frac{\partial^{p_0+\dots+p_d}u}{\partial^{p_0}t\,\partial^{p_1}x_1\cdots\partial^{p_d}x_d}+\dots-g=0$$

SE TUTTI I COEFFICIENTI $a, b, c, \dots \gamma, \dots$ NON DIPENDONO DA u MA SOLO DA x, t $\Rightarrow$ PDE LINEARE.

BUONA PARTE DEL CORSO SU PDE LINEARI.

- PDE <u>SEMILINEARE</u>

$$P(u,g)=\sum_{p_0+\dots+p_d=k}\gamma_{\alpha(p_0,\dots p_d)}(x,t)\,\frac{\partial^{p_0+\dots+p_d}u}{\partial^{p_0}t\;\partial^{p_d}x_d}$$

$$+\,Q(u,g)$$

Q è PDE DI ORDINE $k-1$

- PDE QUASI-LINEARI

$$P(u,g)=\sum_{p_0+\dots+p_d=k}\gamma_{\alpha(p_0,\dots p_d)}(t,x,u,\nabla u,\dots\nabla^{k-1}u)\cdot\frac{\partial^{p_0+\dots+p_d}u}{\partial^{p_0}t\;\partial^{p_d}x_d}$$

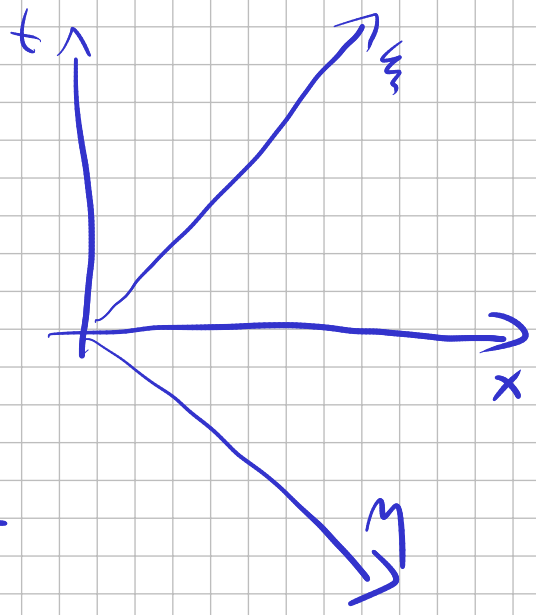
$$+\,Q(u,g)$$

Q ORDINE $k-1$

- ALTRIMENTI TOTALMENTE NON-LINEARE

$$u_t(x,t) + u_x(t,x) = 0$$

$$(x,t) \to (\xi,\eta)$$



$$\xi(x,t) = x+t \qquad \eta = x-t$$

$$x(\xi,\eta) = \frac{\xi+\eta}{2} \qquad t = \frac{\xi-\eta}{2}$$

$$v(\xi,\eta) := u(x(\xi,\eta), t(\xi,\eta))$$

$$u_t + u_x = 0$$

$$\xi_x = 1 \qquad \eta_x = 1$$

$$\xi_t = 1 \qquad \eta_t = -1$$

$$\boxed{u_x =} \; v_\xi \cdot \xi_x + v_\eta \cdot \eta_x = v_\xi + v_\eta$$

$$u(x(\xi,\eta), t(\xi,\eta))$$

$$u_t = v_\xi \xi_t + v_\eta \eta_t = v_\xi - v_\eta$$

$$0 = u_t + u_x = v_\xi + \cancel{v_\eta} + v_\xi - \cancel{v_\eta} = 2 v_\xi = 0$$

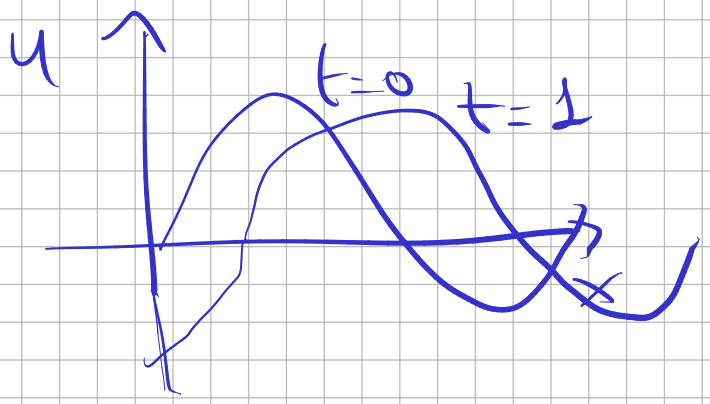
$$v_\xi = 0 \to \text{ODE}$$

$$\Rightarrow v(\xi,\eta) = f(\eta) \qquad f \text{ arbitraria}$$

$$= f(x-t)$$

PER ESEMPIO

$\sin(x-t)$ È UNA SOLUZIONE



- LINEE CARATTERISTICHE SONO QUELLE DOVE LA SOLUZ. HA LO STESSO VALORE

es. $x_{x_0}(t) = x_0 + t$

GENERALIZZIAMO

PDE LINEARE 1° ORDINE

$$a(t,x)\,\partial_t u + b(t,x)\,\partial_x u + c\,u(x,t) = g(x,t)$$

$$(x,t) \in \mathbb{R}^2$$

CERCHIAMO UNA TRASFORMAZIONE

$$(x,t) \xrightarrow{T} (\xi, \eta) \qquad |JT| \neq 0 \qquad \begin{vmatrix} \partial_x \eta & \partial_t \eta \\ \partial_x \xi & \partial_t \xi \end{vmatrix} \neq 0$$

$$u = v$$

$$u_t = v_\xi \xi_t + v_\eta \eta_t$$

$$u_x = v_\xi \xi_x + v_\eta \eta_x$$

$$a \cdot (\xi_t v_\xi + \eta_t v_\eta) + b(\xi_x v_\xi + \eta_x v_\eta)$$

$$+ c \cdot v = g$$

$$(a\xi_t + b\xi_x)V_\xi + \underbrace{(a\eta_t + b\eta_x)}_{\overset{!}{=}0} V_\eta + cV = g$$

OBIETTIVO TROVARE η : $a\eta_t + b\eta_x = 0$

$\Rightarrow$ PDE SI SEMPLIFICA

$$V_\xi + \underbrace{\frac{c}{a\xi_t + b\xi_x}}_{\text{NOTI}} \cdot V = \underbrace{\frac{g}{a\xi_t + b\xi_x}}_{\text{NOTO}}$$

ODE ☺

(1) $a\eta_t + b\eta_x \overset{!}{=} 0$ TROVARE $\eta(x,t)$

$\eta(t, x(t)) = \eta_0$ $\qquad x(t)$?

$$\rightarrow \frac{d}{dt}\eta(t, x(t)) = 0 = \partial_t \eta(t, x(t)) + \partial_x \eta(t, x(t)) \cdot \frac{\partial x}{\partial t}$$

$$\Rightarrow \frac{\partial_t \eta(t, x(t))}{\partial_x \eta(t, x(t))} = -\frac{\partial x(t)}{\partial t}$$

$$(1) \Leftrightarrow \frac{\eta_t}{\eta_x} = -\frac{b}{a} \qquad \frac{\partial x(t)}{\partial t} = \frac{b}{a} \qquad \boxed{\text{ODE}}$$

$\Rightarrow x(t) \quad \forall x_0 \quad \Rightarrow \quad \eta(t, x(t)) = \eta_0$

INVERTI PER η_0

$(x,t) \to (\xi, \eta)$

ESEMPIO

$$x \cdot u_t - t \cdot u_x = 1$$

$$x(v_\eta \eta_t + v_\xi \xi_t) - t(v_\eta \eta_x + v_\xi \xi_x) = 1$$

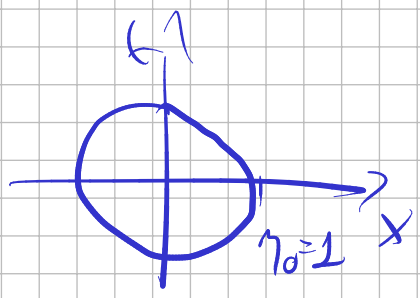
$$\underbrace{(x\eta_t - t\eta_x)}_{=0 \quad (2)} v_\eta + \boxed{(x\xi_t - t\xi_x) v_\xi = 1}$$

$x(t)$: $\quad \eta(t, x(t)) = \eta_0 \qquad \frac{d\eta(t, x(t))}{dt} = 0$

$$0 = \eta_t + \eta_x \frac{dx}{dt} = 0 \qquad \boxed{\frac{dx}{dt} = -\frac{\eta_t}{\eta_x}}$$

$$(2) \Rightarrow \frac{\eta_t}{\eta_x} = \frac{t}{x} \qquad \frac{dx}{dt} = -\frac{t}{x} \quad \text{ODE}$$

$$\int_{x_0}^{x} x\,dx = \int_{t_0}^{t} -t\,dt$$



$$\frac{x^2}{2} - \frac{x_0^2}{2} = -\frac{t^2}{2} + \frac{t_0^2}{2}$$

$$x^2 = -t^2 + \eta_0^2 \qquad x(t) = \pm\sqrt{\eta_0^2 - t^2}$$

$$\Downarrow$$

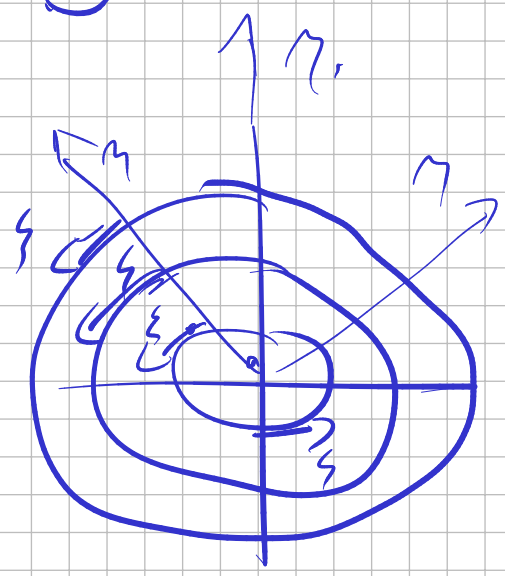
$$\eta(t,x) := \sqrt{x^2 + t^2}$$

CONTROLLO CHE $\quad x\eta_t - t\eta_x = 0$

$$\eta_t = \frac{\cancel{2}\,t}{\cancel{2}\sqrt{x^2+t^2}} \qquad \eta_x = \frac{x}{\sqrt{x^2+t^2}}$$

$$\frac{xt}{\sqrt{x^2+t^2}} - \frac{tx}{\sqrt{x^2+t^2}} = 0 \quad \checkmark \qquad :)$$

$$\eta \checkmark = \sqrt{x^2+t^2} \qquad \xi$$



$$\xi := \arctan\left(\frac{x}{t}\right)$$

$$\xi_t = \frac{-x}{x^2+t^2} \qquad \xi_x = \frac{t}{x^2+t^2}$$

$$t = \eta \cdot \cos(\xi) \qquad x = \eta \sin(\xi)$$

$$(x\xi_t - t\xi_x)\, v_\xi = 1$$

$$\left(\frac{-x^2}{x^2+t^2} - \frac{t^2}{x^2+y^2}\right) v_\xi = 1$$

$$-v_\xi = 1 \qquad v_\xi = -1 \qquad \forall f \text{ ARBITRARIA}$$

$$v(\xi,\eta) = -\xi + f(\eta)$$

$$V(\xi,\eta) = -\xi + f(\eta) = -\arctan\left(\frac{x}{t}\right) + f\left(\sqrt{x^2+t^2}\right)$$

- SECONDO ORDINE PDE LINEARI

$$P(u,g) = A\,u_{xx} + B\,u_{xy} + C\,u_{yy} + D\,u_x + E\,u_y + F\,u - g = 0$$

$\boxed{\text{DEF}}$

DETERMINANTE DELLA PDE 2° ORDINE LINEARE
DISCRIMINANTE

$$\Delta := B^2 - 4AC$$

- $\Delta > 0$ IPERBOLICA
- $\Delta = 0$ PARABOLICA
- $\Delta < 0$ ELLITTICA

ESEMPI
- $u_{tt} - \gamma^2 u_{xx} = 0$ ONDE

$$\Delta = 4\gamma^2 > 0$$ IPERBOLICA

- PARABOLICO : CALORE

$$\partial_t u - \gamma\,\partial_{xx} u = 0 \qquad c > 0$$

$$\Delta = 0 - 0 = 0$$

- ELLITTICO EQ. DI POISSON

$$-\gamma\, \partial_{xx} u - \gamma\, \partial_{yy} u = f$$

$$-\gamma \Delta u = f \qquad \gamma > 0$$

$$\Delta = 0 - 4\gamma^2 = -4\gamma^2 < 0$$

- TRICOMI

$$y\, u_{xx} + u_{yy} = 0$$

$$\Delta = 0 - 4y = -4y \quad \begin{cases} y > 0 & \text{ELLITTICA} \\ y = 0 & \text{PARABOLI} \\ y < 0 & \text{IPERB} \end{cases}$$

$\boxed{\text{TEOREMA}}$

- CAMBIANDO LE VARIABILI IL SEGNO DI Δ NON CAMBIA!

$$(x,y) \to (\xi, \eta)$$

$$u_{xx} = \alpha^2 u_{\xi\xi} + 2\alpha\beta\, u_{\xi\eta} + \beta^2 u_{\eta\eta}$$